## ASCII MASTER FLOW — PANEL 1/12: Population balance equation

POPULATION AT T2 = population at T1 + births - deaths +/- migration
CENSUS -> time-specific snapshot, not a continuous live counter
GROWTH RATE -> outcome of all three components together
TRAP -> birth rate alone cannot explain total population change

MUST REMEMBER: Population geography separates stock, structure, distribution, density, growth, fertility, mortality and mobility; demographic transition moves from high birth/high death through mortality-led growth and fertility decline toward low rates, with a possible ageing or below-replacement phase rather than one universal timetable.

MUST REMEMBER: Population geography separates stock, structure, distribution, density, growth, fertility, mortality and mobility; demographic transition moves from high birth/high death through mortality-led growth and fertility decline toward low rates, with a possible ageing or below-replacement phase rather than one universal timetable.

MUST REMEMBER: Population geography separates stock, structure, distribution, density, growth, fertility, mortality and mobility; demographic transition moves from high birth/high death through mortality-led growth and fertility decline toward low rates, with a possible ageing or below-replacement phase rather than one universal timetable.

MUST REMEMBER: Population geography separates stock, structure, distribution, density, growth, fertility, mortality and mobility; demographic transition moves from high birth/high death through mortality-led growth and fertility decline toward low rates, with a possible ageing or below-replacement phase rather than one universal timetable.

## ASCII MASTER FLOW — PANEL 2/12: Demographic transition spine

```
STAGE 1 -> high birth, high death, unstable growth
STAGE 2 -> death rate falls first, birth stays high
STAGE 3 -> fertility falls, growth slows
STAGE 4/5 -> low rates, then ageing or decline in some societies
```

## ASCII MASTER FLOW — PANEL 3/12: Population pyramid reading

EXPANSIVE -> broad base, youthful age structure, high fertility
STATIONARY -> moderated base and fuller middle, slower growth
CONSTRICTIVE -> narrow base and wider upper ages, ageing trend
READING RULE -> shape, dependency and labour supply move together

## ASCII MASTER FLOW — PANEL 4/12: Dependency to dividend fork

```
WORKING-AGE SHARE RISES -> dependency ratio can fall
IF health + education + skills + jobs improve -> dividend window
IF labour absorption fails -> unemployment and informality rise
VERDICT -> demography gives potential, institutions decide the outcome
```

## ASCII MASTER FLOW — PANEL 5/12: Population-resource triangle

UNDER-POPULATION -> too few people to use resources fully
OVER-POPULATION -> pressure lowers living standards and employment quality
OPTIMUM POPULATION -> relative balance of people, resources and technology
SHIFT RULE -> technology, trade and institutions can move the optimum

## ASCII MASTER FLOW — PANEL 6/12: India demographic timeline

```
1901-1921 -> stagnation under high birth and high death rates
1921-1951 -> steady growth after the turning point
1951-1981 -> population explosion as mortality falls sharply
1981-2011 -> slowing growth as fertility decline deepens
```

## ASCII MASTER FLOW — PANEL 7/12: India data-source firewall

```
CENSUS 2011 -> 1.21 billion total; sex ratio 943; literacy 74.04%
SRS 2022 / NFHS-5 -> TFR 2.0; fertility anchor, not census count
UN WPP 2024 -> mid-2024 estimate about 1.45 billion projection
RULE -> keep instrument, date and status label attached to every figure
```

## ASCII MASTER FLOW — PANEL 8/12: Kerala-EAG contrast rail

```
KERALA -> early fertility decline, higher literacy, later-ageing pressures
EAG STATES -> more youthful structures and slower transition timing
NATIONAL AVERAGE -> hides staggered state-level transition stages
EXAM USE -> India contains early, middle and late transition spaces together
```

## ASCII MASTER FLOW — PANEL 9/12: Ageing with population momentum

```
REPLACEMENT-LEVEL TFR -> does not stop growth immediately
YOUNG COHORT BULGE -> population momentum keeps totals rising
EARLY-TRANSITION STATES -> old-age burden appears sooner
LATE-TRANSITION STATES -> youth bulge and migration pressure persist longer
```

## ASCII MASTER FLOW — PANEL 10/12: Sex ratio interpretation rule

SEX RATIO -> shaped by births, mortality and migration together
LABOUR MIGRATION -> can skew all-age ratios without changing births
CHILD SEX RATIO -> isolates a different part of the demographic problem
SAFE ANSWER -> define the measure before comparing regions

CLOSE DISTINCTION: Crude birth/death rates, TFR, replacement fertility, dependency ratio, sex ratio, life expectancy and population pyramid answer different questions; demographic dividend is a conditional age-structure opportunity, not an automatic growth bonus, and Census 2011, SRS, NFHS and UN estimates retain different reference dates and methods.

CLOSE DISTINCTION: Crude birth/death rates, TFR, replacement fertility, dependency ratio, sex ratio, life expectancy and population pyramid answer different questions; demographic dividend is a conditional age-structure opportunity, not an automatic growth bonus, and Census 2011, SRS, NFHS and UN estimates retain different reference dates and methods.

CLOSE DISTINCTION: Crude birth/death rates, TFR, replacement fertility, dependency ratio, sex ratio, life expectancy and population pyramid answer different questions; demographic dividend is a conditional age-structure opportunity, not an automatic growth bonus, and Census 2011, SRS, NFHS and UN estimates retain different reference dates and methods.

CLOSE DISTINCTION: Crude birth/death rates, TFR, replacement fertility, dependency ratio, sex ratio, life expectancy and population pyramid answer different questions; demographic dividend is a conditional age-structure opportunity, not an automatic growth bonus, and Census 2011, SRS, NFHS and UN estimates retain different reference dates and methods.

## ASCII MASTER FLOW — PANEL 11/12: World divergence under one model

```
SUB-SAHARAN AFRICA -> youthful, high-growth structures remain common
EUROPE / EAST ASIA -> low fertility, ageing and labour shortage concerns
INDIA -> dividend window plus internal divergence at the same time
LESSON -> one transition model yields very different policy problems
EVIDENCE LIMIT: India must be mapped through Kerala/Tamil Nadu ageing and low fertility,
  EAG-state youth and higher fertility, urban-rural and gender contrasts, and
  migration-adjusted state patterns; transition theory is descriptive, while policy, health,
  education, labour absorption and gender agency explain divergent pathways.
```

```
EVIDENCE LIMIT: India must be mapped through Kerala/Tamil Nadu ageing and low fertility,
  EAG-state youth and higher fertility, urban-rural and gender contrasts, and
  migration-adjusted state patterns; transition theory is descriptive, while policy, health,
  education, labour absorption and gender agency explain divergent pathways.
EVIDENCE LIMIT: India must be mapped through Kerala/Tamil Nadu ageing and low fertility,
  EAG-state youth and higher fertility, urban-rural and gender contrasts, and
  migration-adjusted state patterns; transition theory is descriptive, while policy, health,
  education, labour absorption and gender agency explain divergent pathways.
EVIDENCE LIMIT: India must be mapped through Kerala/Tamil Nadu ageing and low fertility,
  EAG-state youth and higher fertility, urban-rural and gender contrasts, and
  migration-adjusted state patterns; transition theory is descriptive, while policy, health,
  education, labour absorption and gender agency explain divergent pathways.
```

## ASCII MASTER FLOW — PANEL 12/12: Population answer spine

```
DEFINE -> indicator, census or survey, and what exactly is being measured
ORDER -> transition stage, pyramid shape and dependency condition
LOCATE -> world contrast plus Kerala-EAG or other state divergence
QUALIFY -> projection versus census, dividend versus momentum, no fabricated PYQ
```